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The baryonic-to-halo mass relation in ΛCDM cosmology: a two-branch law from the energy cascade of dark matter, tested with SPARC

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ABSTRACT

Galaxies hold only about one tenth of the cosmic baryons, and the fraction of a halo's baryons that a galaxy retains peaks near a halo mass of $10^{12}\,\mathrm{M}_\odot$. These facts are usually attributed to feedback. We ask whether part of this structure follows from the dynamics of dark matter itself. Dark matter flow carries kinetic energy from small to large mass scales at a rate $\varepsilon_u = -\frac{3}{2}u^2/t \approx -4.6\times10^{-7}\,\mathrm{m^2\,s^{-3}}$, where u is the velocity dispersion of all particles. If baryons and dark matter in a halo share this rate, and the baryonic Tully–Fisher relation (BTFR) $v_f^4 = Gm_b a_0$ holds, two regimes follow. In small haloes the baryons move with the dark matter, $v_f \propto v_{\rm cir}$ and $m_b \propto m_h^{4/3}$. In large haloes the baryons carry the cascade with their own rotation, $v_f \propto v_{\rm cir}^{1/3}$ and $m_b \propto m_h^{4/9}$. The two branches meet at $m_{hc} \approx 1.34\times10^{12}\,\mathrm{M}_\odot$, where the baryon-to-halo mass ratio peaks at $A \approx 0.076$. We test this on the 123 SPARC galaxies with a measured flat rotation velocity, using published halo fits. A break is required by the data ($\Delta\mathrm{BIC} = 72$ over a single power law). It lies at $M_{200} = 10^{11.95}\,\mathrm{M}_\odot$ and $V_{200} = 147\,\mathrm{km\,s^{-1}}$, against the predicted $10^{12.12}\,\mathrm{M}_\odot$ and $143\,\mathrm{km\,s^{-1}}$. Above the break the slope is 0.21 (68 per cent range -0.04 to 0.34; predicted 0.44). Below it the slope is 1.55 (1.44–1.62), steeper than the predicted 4/3. The halo masses carry an abundance-matching prior. Fits without that prior, restricted to galaxies with well-measured halo masses, keep the break ($10^{12.02}\,\mathrm{M}_\odot$) and are consistent with both predicted slopes. Averaged over the halo mass function, the relation gives a mean baryon-to-halo mass ratio of 0.024 and puts 7.6 per cent of all baryons in galaxies if only haloes above $10^{12}\,\mathrm{M}_\odot$ host them. The fraction rises to 12 per cent if haloes down to $10^{10}\,\mathrm{M}_\odot$ do. The low-mass slope and the dependence on the halo-mass prior are the main open points.

Key words: dark matter – galaxies: haloes – galaxies: kinematics and dynamics – galaxies: formation – methods: analytical

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1 INTRODUCTION

The connection between galaxies and their dark matter haloes is one of the main empirical results of the last two decades [1]. Abundance matching links the stellar mass of a galaxy to the mass of its halo. The ratio M_*/M_h peaks at a few per cent near $M_h \approx 10^{12}\,\mathrm{M}_\odot$ and falls steeply on both sides [2, 3, 4, 5, 6]. The standard explanation is feedback. Supernovae remove gas from shallow potential wells [7], and active nuclei stop the cooling of gas in massive haloes [8]. The location of the peak and the two slopes then depend on the strength of these processes.

Four further observations frame the question of this paper.

(i) *Missing baryons.* Galaxies contain only about one tenth of the cosmic baryons [9, 10, 11, 12]. The baryon fraction of galaxies is far below the cosmic value at all masses [13].

(ii) *The baryonic Tully–Fisher relation.* The total baryonic mass of a rotating galaxy follows $m_b \propto v_f^4$, with v_f the flat rotation velocity, and has very small scatter [14, 15]. Its normalisation defines an acceleration $a_0 \approx 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$, the constant of modified Newtonian dynamics [16].

(iii) *Massive discs.* Rotation curves of the most massive spiral galaxies imply that they have turned most of their halo baryons into stars. The efficiency shows no decline above $10^{12}\,\mathrm{M}_\odot$ [17].

(iv) *Halo masses from rotation curves.* Halo masses of individual discs are only weakly constrained by their rotation curves. Published fits therefore use priors that include the abundance-matching relation itself [18, 19].

Dark matter in the cold dark matter model is collisionless and interacts only through gravity. On large scales it behaves as a fluid. Its flow resembles turbulence in some respects and differs in others [20]. Haloes merge and grow, which carries mass from small to large mass scales (an inverse mass cascade). The kinetic energy is carried along at a rate that does not depend on mass scale [21, 22]. This paper asks whether the rate of this cascade, together with the BTFR,

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fixes the baryonic-to-halo mass relation (BHMR). The relation is m_b/m_h against m_h, with m_b the mass in stars and cold gas.

The derivation (Sections 2–3) gives two branches with fixed slopes and a pivot mass set by three quantities: the velocity dispersion u, the cascade rate ε_u and G. We test the slopes and the pivot against the SPARC rotation curves with halo fits made with and without abundance-matching priors (Section 4). We then derive the baryon budget of haloes and its dependence on the smallest halo that hosts a galaxy (Section 5). Section 6 lists what the model explains and what it does not.

2 ENERGY CASCADE IN DARK MATTER FLOW

We use the SCDM simulation of the Virgo consortium [23, 24]: 256^3 particles of $m_p = 2.27\times10^{11}\,\mathrm{M}_\odot/h$ in a box of 239.5 Mpc/h, $\Omega_m = 1$, $h = 0.5$. Haloes are found with a friends-of-friends algorithm (comoving linking length 0.2 Mpc/h) and grouped by their number of particles n_p. The group statistics are public [25].

Mass cascade. Let $m_g(n_p,t)$ be the mass in haloes of n_p particles. The mass flux across n_p is

$$\Pi_m(n_p,t) = -\sum_{n>n_p}\frac{\partial m_g(n,t)}{\partial t}, \tag{1}$$

the rate at which mass leaves all groups smaller than n_p for larger ones. In the propagation range (small n_p) the flux does not depend on n_p:

$$\Pi_m = \varepsilon_m = -\frac{M_h H}{2} = -\frac{M_h}{3t}, \tag{2}$$

where M_h is the total mass in haloes and $t = 2/(3H)$. Fig. 1(a) shows the measured flux; at $0 < z < 0.1$ its plateau is 1.07 times Eq. (2).

Energy cascade. The kinetic energy of halo particles splits into the motion of haloes (σ_h^2, the one-dimensional halo velocity dispersion) and the motion inside haloes (σ_v^2). Replacing $\partial m_g/\partial t$ by $\sigma^2\partial m_g/\partial t$ in Eq. (1) gives the energy fluxes Π_{kh} and Π_{kv} (Fig. 1b,c). They also have plateaus. At $z\to0$ the plateau of Π_{kv} is 0.93 times $5\varepsilon_m\langle\sigma_v^2\rangle$ and that of Π_{kh} is 0.55 times $\varepsilon_m\langle\sigma_h^2\rangle$. The energy flux follows the mass flux, with coefficients set by how fast the specific energies of haloes change in time [22]. The kinetic energy of the whole system grows at the characteristic rate

$$\varepsilon_u = -\frac{3}{2}\frac{u^2}{t} = -\frac{9}{4}Hu^2 = -4.6\times10^{-7}\,\mathrm{m^2\,s^{-3}}, \tag{3}$$

with $u_0 = 354.6\,\mathrm{km\,s^{-1}}$ the one-dimensional velocity dispersion of all particles at $z = 0$ and $t_0 = 13.04\,\mathrm{Gyr}$. Equation (3) is a scale: in this simulation u^2 grows somewhat slower than t [26], and $\langle\sigma^2\rangle M_h/M$ of halo particles is $0.75\,u^2$ at $z = 0$.

3 TWO-FLUID MODEL AND THE TWO-BRANCH RELATION

A halo of mass m_h and radius r_h has the circular velocity

$$v_{\rm cir}^2 = \frac{Gm_h}{r_h} = \frac{\Delta_c}{2}(Hr_h)^2, \qquad m_h = \frac{4}{3}\pi r_h^3\Delta_c\bar\rho, \tag{4}$$

with Δ_c the mean density of the halo in units of the background density $\bar\rho$. For $\Delta_c = 18\pi^2$ the orbital time $2\pi r_h/v_{\rm cir}$ equals the age t, and Eq. (3) becomes $\varepsilon_u = \frac{3}{2}u^2/(2\pi r_h/v_{\rm cir})$. The cascade carries the specific energy $\frac{3}{2}u^2$ once per orbit.

We treat the baryons and the dark matter of a halo as two fluids that exchange energy only through gravity (Fig. 2) and assume that both carry the cascade at the same rate ε_u. The baryons are described by the flat rotation velocity v_f, which obeys the BTFR,

$$v_f^4 = Gm_b a_0. \tag{5}$$

Small haloes. Haloes with masses well below the characteristic mass of the mass function accrete slowly. Their dark matter is close to stable clustering, so the proper flow is nearly incompressible and the baryons are carried with it. The specific energy transferred is that of the dark matter, u^2, over the baryonic orbital time r_h/v_f:

$$\begin{aligned}\varepsilon_u &= -\beta_f\frac{u^2 v_f}{r_h}a^q,\\ v_f &= \frac{9}{4\beta_f}\Big(\frac{2}{\Delta_c}\Big)^{1/3}(Gm_hH)^{1/3}a^{-q}\propto v_{\rm cir},\end{aligned} \tag{6}$$

with a constant β_f and $q = -1/2$, so that v_f of a given halo does not change as $u^2\propto a^{3/2}$ grows.

Large haloes. Massive haloes accrete fast and keep a nearly constant concentration. The two fluids are not locked together, and the energy carried by the baryons is that of their own rotation, v_f^2, over the same time:

$$\begin{aligned}\varepsilon_u &= -\alpha_f\frac{v_f^3}{r_h}a^p,\\ v_f &= \Big(\frac{9}{4\alpha_f}\Big)^{1/3}\Big(\frac{2}{\Delta_c}\Big)^{1/9}u^{2/3}(Gm_hH)^{1/9}a^{-p/3}\propto v_{\rm cir}^{1/3}.\end{aligned} \tag{7}$$

The two branches. With Eq. (5),

$$m_b\propto m_h^{4/3}\ (m_h<m_{hc}), \qquad m_b\propto m_h^{4/9}\ (m_h>m_{hc}), \tag{8}$$

so m_b/m_h rises as $m_h^{1/3}$ and falls as $m_h^{-5/9}$. The branches meet where Eqs. (6) and (7) give the same v_f:

$$v_{fc} = u\sqrt{\beta_f/\alpha_f}, \qquad m_{hc} = \Big(\frac{\beta_f^3}{\alpha_f}\Big)^{3/2}\frac{8\Delta_c}{81}\frac{u^5}{G|\varepsilon_u|}a^{\frac{3}{2}(3q-p)}. \tag{9}$$

The pivot mass is set by u, ε_u and G; β_f and α_f are dimensionless. The peak ratio is

$$A = \frac{m_{bc}}{m_{hc}} = \frac{2/\Delta_c}{\alpha_f^{1/2}\beta_f^{5/2}}\gamma_f\,a^{-(5q+p)/2}, \qquad \gamma_f = \frac{|\varepsilon_u|^3}{H^2u^5a_0}. \tag{10}$$

A double power law joins the branches,

$$\frac{m_b}{m_h} = 2^{1/m}A\left[\Big(\frac{m_h}{m_{hc}}\Big)^{-m/3} + \Big(\frac{m_h}{m_{hc}}\Big)^{5m/9}\right]^{-1/m}. \tag{11}$$

Constants. Table 1 lists the values. The calibration of $\beta_f = 0.16$ and $\alpha_f = 0.5$ against SPARC uses $\Delta_c = 200$ and the empirical a_0. Written with $a_0 = -(\Delta_c/2)\varepsilon_u/u$, the relation implied by $\varepsilon_u = -a_0u/(3\pi)^2$, the model is closed only for $\Delta_c = 18\pi^2$. With $\Delta_c = 18\pi^2$, $a_0 = 1.15\times10^{-10}\,\mathrm{m\,s^{-2}}$ and $\gamma_f = (3/4\pi)^2 = 0.057$, the peak moves to $A = 0.089$ at $m_{hc} = 1.20\times10^{12}\,\mathrm{M}_\odot$. Both sets are shown because the observed galaxies do not fix Δ_c within the model.

4 TESTS WITH SPARC GALAXIES

Data. SPARC provides rotation curves and mass models of 175 late-type galaxies [27]. The BTFR sample of Lelli *et al.* [15] lists the baryonic mass m_b (stars and cold gas) of 153 galaxies and a flat velocity v_f for 123 of them; we use these 123. Halo masses M_{200} and circular velocities V_{200} (density 200 times critical, $H_0 =$

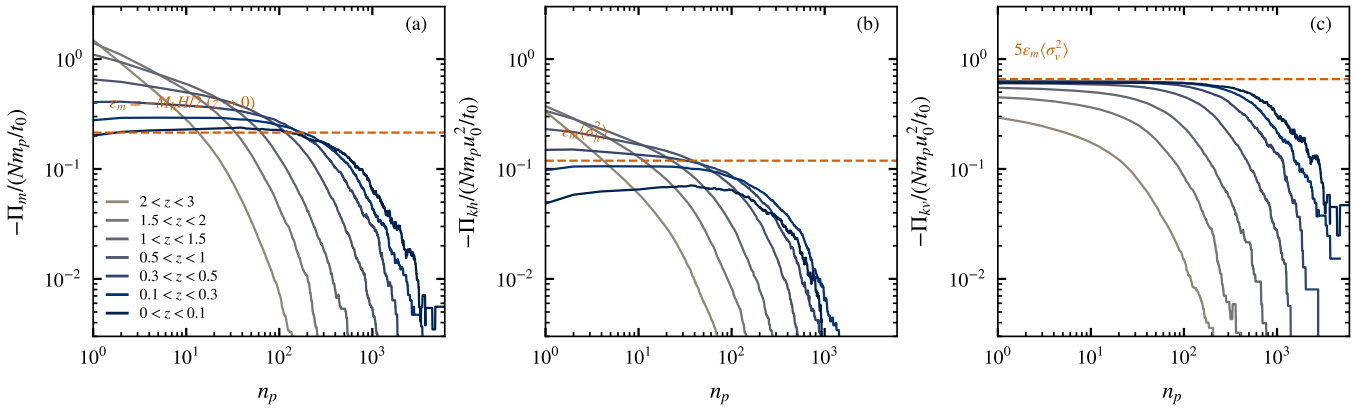


Figure 1. Flux functions of the halo-mediated cascade in the Virgo SCDM simulation for seven pairs of snapshots (colours). (a) Mass flux, Eq. (1); (b) flux of halo kinetic energy σ_h^2; (c) flux of the kinetic energy inside haloes σ_v^2. Dashed lines: closed forms at $z \to 0$. Each flux is constant in the propagation range of small haloes and falls where the cascaded mass is deposited in the largest haloes.

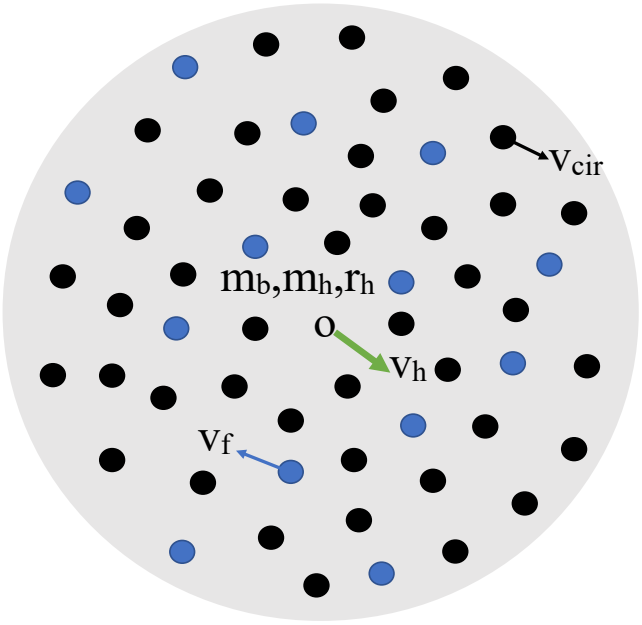


Figure 2. Two-fluid model of a halo of mass m_h and radius r_h. The baryons (mass m_b, typical speed v_f) and the dark matter (mass m_h, typical speed $v_{\rm cir}$) move inside the same halo and interact only through gravity.

Table 1. Constants and predictions of the two-branch relation at $z = 0$. $u_0 = 354.6\,\mathrm{km\,s^{-1}}$, $\varepsilon_u = 4.6 \times 10^{-7}\,\mathrm{m^2\,s^{-3}}$, $H_0 = 1.62 \times 10^{-18}\,\mathrm{s^{-1}}$, $\beta_f = 0.16$, $\alpha_f = 0.5$, $q = -1/2$, $p = 7/4$. Masses in $\mathrm{M}_\odot$.

	$\Delta_c = 200$, $a_0 = 1.2 \times 10^{-10}$	$\Delta_c = 18\pi^2$, a_0 from ε_u
v_{fc} [km s^{-1}]	201	201
v_{cc} [km s^{-1}]	143	134
r_{hc} [kpc]	285	285
m_{hc}	1.34×10^{12}	1.20×10^{12}
m_{bc}	1.02×10^{11}	1.06×10^{11}
A	0.076	0.089
mean m_b/m_h of haloes	0.024	0.027

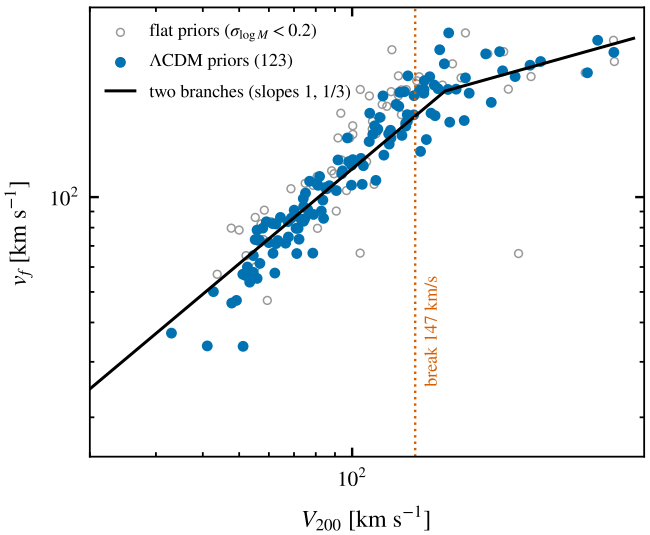


Figure 3. Flat rotation velocity against halo circular velocity for SPARC galaxies (NFW fits; filled: ΛCDM priors, open: flat priors). Black: the two branches, slopes 1 and 1/3, with the break fitted. The dotted line marks the free break of the broken power law.

73 km s^{-1} Mpc^{-1}) come from the rotation-curve fits of Li *et al.* [18]. Their 'ΛCDM' fits impose the concentration–mass relation and the stellar-to-halo mass relation of Moster *et al.* [3] as priors; their 'flat' fits do not. For the flat fits we keep galaxies with $\sigma(\log M_{200}) < 0.2$ (67 NFW, 48 Einasto). All fits are least squares in log space, for a single power law and for a continuous broken power law with a free break. Uncertainties are 16th–84th percentiles of 1000 bootstrap re-samplings of the galaxies.

Velocities. The data require two branches (Fig. 3): a broken power law is preferred over a single one by $\Delta\mathrm{BIC} = 64$. The break lies at $V_{200} = 147\,\mathrm{km\,s^{-1}}$, near the predicted $v_{cc} = 143\,\mathrm{km\,s^{-1}}$. Above it the slope is 0.32 (0.15–0.40), consistent with 1/3. Below it the slope is 1.23 (1.17–1.28), steeper than 1. The median v_f/V_{200} is 1.21. On the small-halo branch the model gives $v_f/v_{\rm cir} = 9/(4\beta_f\sqrt{\Delta_c/2}) = 1.41$, or 1.24 after converting the model radius to R_{200}.

Masses. The BHMR also has two branches (Fig. 4). With NFW

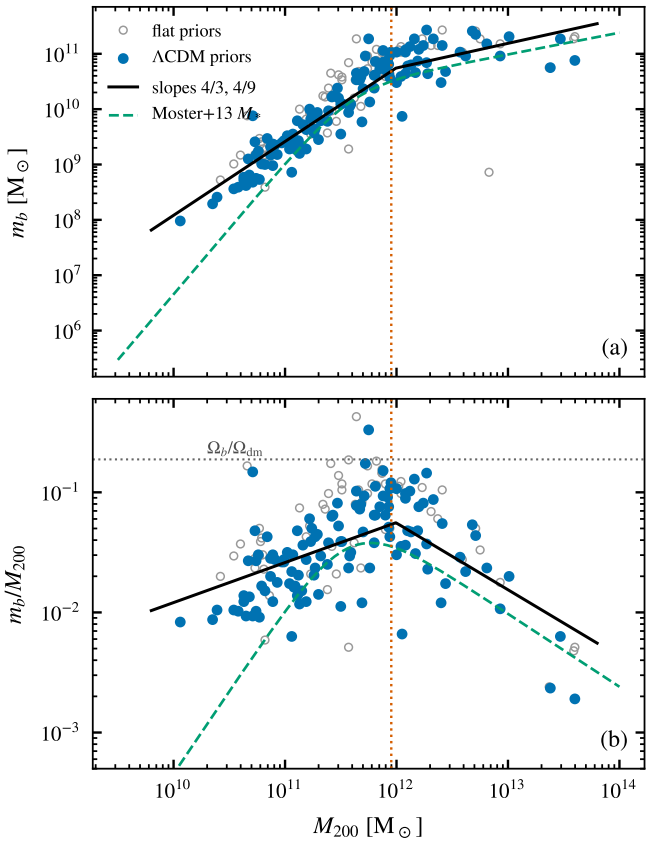


Figure 4. (a) Baryonic mass and (b) baryon-to-halo mass ratio against halo mass for SPARC galaxies. Filled: ΛCDM priors, open: flat priors (NFW). Black: the two branches, slopes 4/3 and 4/9, with the break fitted. Green dashed: the stellar-to-halo relation of Moster *et al.* [3], part of the ΛCDM priors. Dotted horizontal line: cosmic baryon-to-dark-matter ratio.

halo masses and ΛCDM priors, the broken power law beats a single one by $\Delta\mathrm{BIC} = 72$. The break is at $\log_{10}(M_{200}/\mathrm{M}_\odot) = 11.95$ (11.88–12.17), against $\log_{10} m_{hc} = 12.12$. The high-mass slope is 0.21 (-0.04–0.34). The bootstrap places a slope of 4/9 or larger at a probability of 0.04. The low-mass slope is 1.55 (1.44–1.62); a slope of 4/3 or smaller has probability 0.005. Fixing both slopes at 4/3 and 4/9 costs $\Delta\mathrm{BIC} = 5.2$. Fig. 5 collects the four sets of halo fits. Einasto haloes give the same result. With flat priors the NFW fits keep the break at $10^{12.02}\,\mathrm{M}_\odot$, with slopes 1.48 and -0.10. For them the fixed model slopes are not disfavoured ($\Delta\mathrm{BIC} = -0.4$). The Einasto flat-prior fits put the break lower, at $10^{11.06}\,\mathrm{M}_\odot$, with only 48 galaxies. The baryon-to-halo ratio stays below the cosmic value $\Omega_b/\Omega_{\rm dm} = 0.188$ for all but 1 galaxy.

Cascade rate of individual galaxies. Each galaxy gives its own rate through Eq. (6) or (7). With the halo radius defined as in the calibration, the median is $4.6\times10^{-7}\,\mathrm{m^2\,s^{-3}}$ and the scatter 0.12 dex. The scatter is the residual of the two-branch relation. It measures how far individual haloes depart from the mean cascade, not a test of the mean, which the calibration of β_f and α_f fixes.

5 HALO BARYON BUDGET

Averaging Eq. (8) over the mass function gives the mean baryon-to-halo mass ratio of all haloes, $A_{bh} = \int f(\nu)\,(m_b/m_h)\,\mathrm{d}\nu$. Here $\nu = (m_h/m_h^*)^{2/3}$, $m_h^* = 4\times10^{13}\,\mathrm{M}_\odot$ at $z = 0$, and $f(\nu) \propto \nu^{q_0/2-1}e^{-\nu/4\eta_0}$ ($q_0 = 0.556$, $\eta_0 = 0.76$) is the double-λ mass function of the simulation [21]. Both branches integrate in closed form (incomplete gamma and generalized exponential integrals), giving $A_{bh} = 0.024$ at $z = 0$. The baryons in haloes, as a fraction of all baryons, are $A_{dh}A_{bh}/0.188$. Here A_{dh} is the fraction of dark matter in haloes and 0.188 the cosmic baryon-to-dark-matter ratio.

A_{dh} depends on the smallest halo counted. In the simulation, haloes of at least two particles ($m_h \geqslant 9.1\times10^{11}\,\mathrm{M}_\odot$) hold 61 per cent of the dark matter. With that limit, 7.6 per cent of all baryons lie in galaxies, close to the 8 per cent measured by Read and Trentham [10]. Extending the same mass function to smaller haloes raises the fraction to 11 per cent for $m_{\rm min} = 10^{11}\,\mathrm{M}_\odot$ and 12 per cent for $10^{10}\,\mathrm{M}_\odot$ (Fig. 6b). The agreement at the two-particle limit is therefore partly a coincidence of the simulation's resolution. The robust statement is that the relation keeps galaxies at 8–13 per cent of the baryons, in line with the measured 7–10 per cent [9, 10, 11].

With the time dependence of Table 1 the peak ratio grows as $A \propto a^{3/8}$ and the pivot mass falls as $m_{hc} \propto a^{-9/8}$. The fraction of baryons in haloes grows from 0.007 at $z = 10$ to 0.076 at $z = 0$, as $a^{0.63}$ for $z \leqslant 1$ (Fig. 6a).

6 DISCUSSION

What the relation explains. One energy rate and the BTFR give a BHMR with a pivot near $10^{12}\,\mathrm{M}_\odot$ and a high-mass slope close to that of abundance matching. The pivot is set by $u^5/(G\varepsilon_u)$, a combination of the dark matter flow alone. SPARC galaxies show the break at the predicted halo mass and circular velocity. Above the break, $v_f \propto V_{200}^{1/3}$ holds within the errors.

What it does not explain. Below the break the data are steeper than 4/3 in mass and steeper than 1 in velocity. The fits with abundance-matching priors give the clearest disagreement, partly because the stellar-to-halo prior is itself steep there (Fig. 4, $M_* \propto M_h^{2.4}$). The fits without that prior bring the slope to 1.48. Gas-rich dwarfs, whose cold gas dominates m_b, are where the coupling assumption of the small-halo branch is weakest.

Dependence on the halo masses. The pivot of the SPARC data near $10^{12}\,\mathrm{M}_\odot$ is not independent of abundance matching when the ΛCDM-prior halo masses are used. The flat-prior NFW fits keep it; the flat-prior Einasto fits do not. Posti *et al.* [17] find no decline of the efficiency of massive spirals with flat priors and a different mass model. The high-mass branch therefore needs halo masses from weak lensing or satellite kinematics for a decisive test.

Predictions. With the time dependence of Table 1 the pivot mass decreases with time ($m_{hc} \propto a^{-9/8}$) and the peak ratio increases ($A \propto a^{3/8}$). Abundance matching finds a peak that moves to lower halo mass at later times [5, 6]. The two exponents can be tested directly with the redshift evolution of the stellar-to-halo relation.

7 CONCLUSIONS

(i) If the baryons and the dark matter of a halo carry the kinetic-energy cascade of dark matter flow at the same rate, $\varepsilon_u \approx -4.6\times10^{-7}\,\mathrm{m^2\,s^{-3}}$, and the BTFR holds, the BHMR has two branches: $m_b \propto m_h^{4/3}$ in small haloes, where the baryons move with the dark matter, and $m_b \propto m_h^{4/9}$ in large haloes, where they carry the cascade with their own rotation.

(ii) The branches meet at $m_{hc} \propto u^5/(G|\varepsilon_u|)$: $1.34\times10^{12}\,\mathrm{M}_\odot$ with

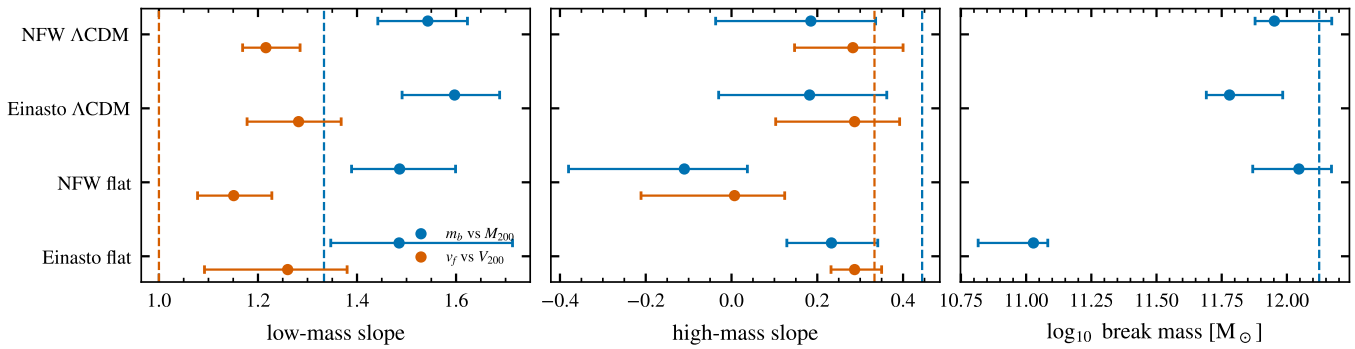


Figure 5. Low-mass slope, high-mass slope and break of the broken power law for four sets of SPARC halo fits (NFW, Einasto; ΛCDM or flat priors). Blue: m_b against M_{200}; red: v_f against V_{200}. Bars: 68 per cent bootstrap ranges. Dashed: predictions of the two-branch relation (4/3, 4/9 and $m_{hc} = 1.33 \times 10^{12}\,\mathrm{M}_\odot$; 1 and 1/3).

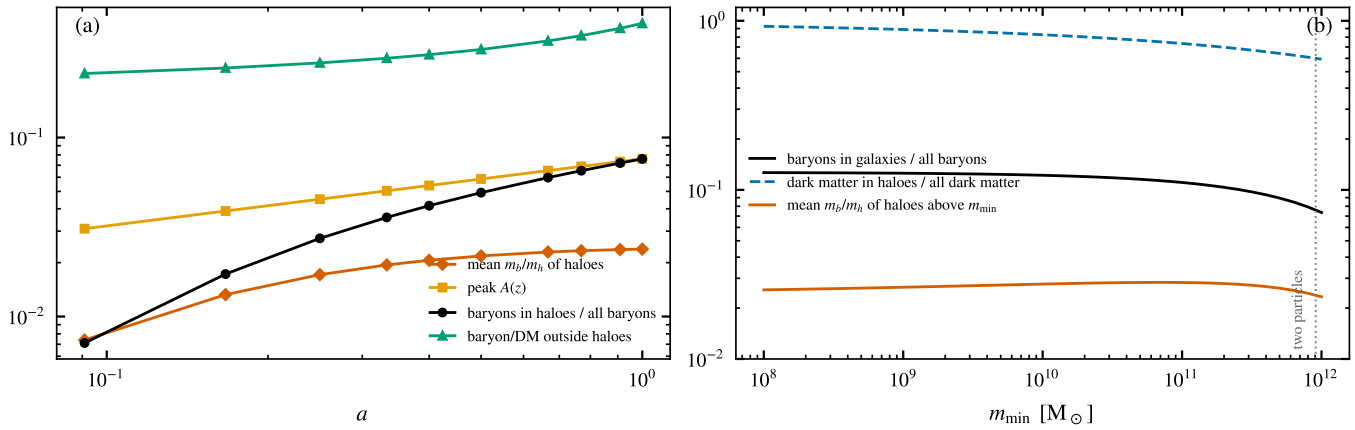


Figure 6. (a) Evolution of the halo baryon budget for the two-branch relation with the time dependence of Table 1: mean m_b/m_h of haloes, peak ratio A, baryons in haloes as a fraction of all baryons, and baryon-to-dark-matter ratio outside haloes. (b) Dependence on the smallest halo that hosts a galaxy, m_{min}, at $z = 0$. The dotted line marks the mass of two simulation particles, the limit of the halo catalogue.

the calibration of SPARC ($\Delta_c = 200$), $1.20 \times 10^{12}\,\mathrm{M}_\odot$ for the self-consistent $\Delta_c = 18\pi^2$. The peak ratio is 0.076–0.089.

(iii) SPARC galaxies require a break ($\Delta\mathrm{BIC} = 72$) at $M_{200} = 10^{11.95}\,\mathrm{M}_\odot$ and $V_{200} = 147\,\mathrm{km\,s^{-1}}$, as predicted. The high-mass slopes agree with 4/9 and 1/3; the low-mass slopes are steeper than 4/3 and 1 (1.55 and 1.23).

(iv) Halo masses with abundance-matching priors make the pivot partly built in. Prior-free fits with well-measured halo masses keep the break and the predicted slopes in NFW haloes but not in Einasto haloes.

(v) Averaged over the halo mass function, the mean baryon-to-halo ratio is 0.024. Galaxies hold 7.6–12 per cent of all baryons, depending on the smallest halo that hosts a galaxy, consistent with the observed census.

DATA AVAILABILITY

The halo-based statistics of the simulation are available on Zenodo [25] and GitHub [28]. The SPARC data and halo fits are public [18, 27]. The code that computes every number and figure, and CSV files of all results, are in the reproducible package that accompanies this paper.

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